

MA1513 Weekly Self Practice Set 1

1. (a) Write down the original linear system with the following augmented matrix. (Use variables x_1, x_2, x_3, x_4)

$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 2 & 5 & -4 & 2 & 15 \\ 3 & 5 & -1 & 3 & 20 \end{array} \right)$$

- (b) Use Gaussian elimination to reduce the augmented matrix in part (a) to the reduced row echelon form

$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

using exactly four elementary row operations. Write down the elementary row operation that you use in each step clearly.

Present your answer in the following format:

$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 2 & 5 & -4 & 2 & 15 \\ 3 & 5 & -1 & 3 & 20 \end{array} \right) \xrightarrow{ERO1} \dots \xrightarrow{ERO2} \dots \xrightarrow{ERO3} \dots \xrightarrow{ERO4} \left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

- (c) Write down the general solution of the linear system in part (a).

2. (a) Write the linear system

$$\begin{cases} 3x + 2y = 6 \\ 2x + 2y = 8 \end{cases}$$

in matrix equation form $\mathbf{Ax} = \mathbf{b}$.

- (b) Find the inverse matrix $\mathbf{A}^{-1}$ of $\mathbf{A}$ in part (a) by hand.
(c) Use the answer in (b) to find the solution of the matrix equation form $\mathbf{Ax} = \mathbf{b}$ in (a). Show your working, and give your answer of $\mathbf{x}$ as a column matrix.